**Project Title:**
Customer Segmentation and Targeting Strategy Using Behavioral Data

This project aims to analyze customer behavior and identify meaningful segments using data science techniques. The main research question is whether customers can be grouped based on their demographic and behavioral characteristics, and how these groups can be used to improve targeting strategies.

The primary dataset will be the Mall Customers Dataset, which is publicly available and includes variables such as age, annual income, and spending score. The dataset will be obtained from Kaggle and accessed using Python. Since the dataset is public, it will be enriched through feature engineering, including creating variables such as normalized spending behavior, income-to-spending ratios, and other derived indicators to better capture customer behavior.

The dataset contains approximately 200 customer observations with multiple numerical attributes. These features allow for clustering and behavioral analysis.

The analysis will begin with data cleaning and preprocessing, followed by exploratory data analysis (EDA) to understand patterns and relationships within the data. Customer segmentation will then be performed using clustering methods such as K-Means and hierarchical clustering. The optimal number of clusters will be determined using methods like the Elbow Method and Silhouette Score. Finally, the identified segments will be interpreted and translated into actionable targeting strategies, such as identifying high-value customers, price-sensitive groups, and low-engagement users.

The expected outcome of this project is to generate clear and interpretable customer segments and provide data-driven recommendations for improving marketing efficiency and targeting strategies.